



— THERMAL OIL SYSTEMS



ENGINEERED HEAT FOR INDUSTRIAL PROCESSES

**High-temperature thermal oil systems
designed for continuous and reliable operation.**

Thermal Oil for Process Energy

In many industrial processes requiring high-temperature heat exchange, conventional systems such as superheated water and steam can be used, but they involve high operating and maintenance costs due to water treatment and complex control systems.

Thermal oil systems provide a reliable and efficient alternative, enabling high operating temperatures at very low system pressures for multiple heat users.

Operating efficiently within the 150 °C – 350 °C range, thermal oil heating systems offer significant advantages. While steam systems require pressures of up to 90 bar at 300 °C, thermal oil systems operate only at the circulation pump pressure. The thermal fluid circulates in closed loops, ensuring long service life with reduced maintenance and operating costs.

Easy operation, short start-up time, and precise temperature control at low operating pressure are the key advantages of thermal oil systems.

**" THERMAL OIL HEATERS OPERATE ONLY AT THE PUMPING
PRESSURE OF THE CIRCULATION SYSTEM "**

Advantages of Thermal Oil Heaters

- High temperature applications @ atmospheric pressure
- Secondary circuit applications allow:
 - **Accurate temperature control over your system ($\pm 1\text{ }^{\circ}\text{C}$)**
 - Different operating temperatures for multiple heat users
- No condensate & corrosion problems
- No water treatment
- Single-point heat generation for multiple heat-consuming units
- Ease of operation and maintenance
- Steam generation is possible via indirect steam generator
- Possibility of integration an **with ORC turbine for Electricity Generation**

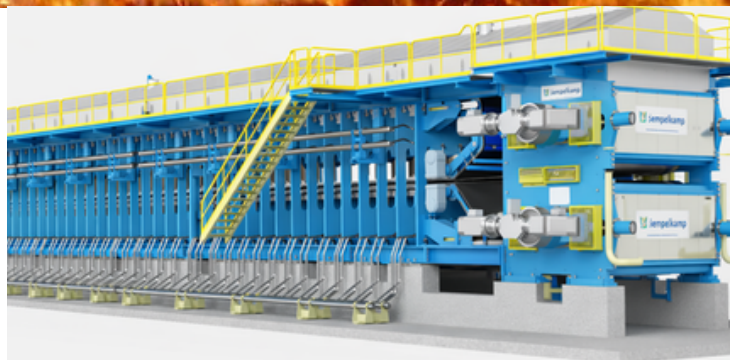
Technical Specifications

- Standard capacities from **0.4 to 20 MW** in vertical / horizontal configuration
- Capacities up to **50 MW** with combined configuration
- Suitable for oil, gas & oil + gas co-fired operations
- Designed in accordance with DIN 4754;
- Coils manufactured from seamless steel tubes in accordance with EN 10216-2 / EN 10220 / EN 10204
- Optimum thermal design ensures efficient heat transfer from flue gases to thermal oil
- Thermal oil flow velocity designed and controlled according to applicable standards
- Continuous flow monitoring prevents overheating, cracking, corrosion, & film formation, extending the service life of thermal oil.

Application Fields of Thermal Oil

Wood & Wood Products Industry

- Hot presses and rolling presses
- Veneer and board drying units
- Impregnation processes
- Process air heating applications



Paper & Cardboard Industry

- Paper and board press heating
- Roller and bath heating
- Felt asphalt coating
- Impregnation processes



Construction Industry

- Heating of material silos
- Concrete defrosting systems
- Brick and ceramic drying
- Refractory material impregnation



Textile Industry

- Yarn drying and curing
- Coating and finishing lines
- Dyeing and bleaching baths
- Climatization applications



Food & Beverage Industry

- Bottle washing systems
- Boiling and frying units
- Pasteurization of milk and juices
- Confectionery production processes



Petroleum & Chemical Industry

- Process tank and pipeline heating
- Asphalt, bitumen, and coal tar systems
- PVC and polymer production processes
- Reactor and autoclave heating



In House Manufacturing

- Full control over quality & manufacturing standards
- Faster delivery with engineering and production
- High flexibility for project-specific designs
- Consistency with international standards

Quality Control Process

- Material traceability (certified supply)
- Welding procedures (WPS, PQR)
- Non-destructive testing (RT, UT, PT, MT)
- Pressure and leakage tests

Product Certifications and Standards

- CE Marking
- ASME U + S + R
- EN 12952 / EN 12953 (European boiler standards)
- PED 2014/68/EU (Pressure Equipment Directive)
- AD 2000 Merkblatt, EN 13445, EN ISO 3834-2
- EAC – (Gost-R) for Russian Federation
- Razresheniye Gospromnadzor for Republic of Belarus

